



Designation: D2487 – 17^{ε1}

Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)¹

This standard is issued under the fixed designation D2487; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

^{ε1} NOTE—Footnote L in Table 1 was editorially corrected in April 2020.

1. Scope*

1.1 This practice describes a system for classifying mineral and organo-mineral soils for engineering purposes based on laboratory determination of particle-size characteristics, liquid limit, and plasticity index and shall be used when precise classification is required.

NOTE 1—Use of this standard will result in a single classification group symbol and group name except when a soil contains 5 to 12 % fines or when the plot of the liquid limit and plasticity index values falls into the crosshatched area of the plasticity chart. In these two cases, a dual symbol is used, for example, GP-GM, CL-ML. When the laboratory test results indicate that the soil is close to another soil classification group, the borderline condition can be indicated with two symbols separated by a slash. The first symbol should be the one based on this standard, for example, CL/CH, GM/SM, SC/CL. Borderline symbols are particularly useful when the liquid limit value of clayey soils is close to 50. These soils can have expansive characteristics and the use of a borderline symbol (CL/CH, CH/CL) will alert the user of the assigned classifications of expansive potential.

1.2 The group symbol portion of this system is based on laboratory tests performed on the portion of a soil sample passing the 3-in. (75-mm) sieve (see Specification E11).

1.3 As a classification system, this standard is limited to naturally occurring soils.

NOTE 2—The group names and symbols used in this test method may be used as a descriptive system applied to such materials as shale, claystone, shells, crushed rock, etc. See Appendix X2.

1.4 This standard is for qualitative application only.

NOTE 3—When quantitative information is required for detailed designs of important structures, this test method must be supplemented by laboratory tests or other quantitative data to determine performance characteristics under expected field conditions.

1.5 This standard is the ASTM version of the Unified Soil Classification System. The basis for the classification scheme

is the Airfield Classification System developed by A. Casagrande in the early 1940s.² It became known as the Unified Soil Classification System when several U.S. Government Agencies adopted a modified version of the Airfield System in 1952.

1.6 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.7 *This practice offers a set of instructions for performing one or more specific operations. This document cannot replace education or experience and should be used in conjunction with professional judgment. Not all aspects of this practice may be applicable in all circumstances. This ASTM standard is not intended to represent or replace the standard of care by which the adequacy of a given professional service must be judged, nor should this document be applied without consideration of a project's many unique aspects. The word "Standard" in the title of this document means only that the document has been approved through the ASTM consensus process.*

1.8 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:³

C117 Test Method for Materials Finer than 75-μm (No. 200) Sieve in Mineral Aggregates by Washing

¹ This practice is under the jurisdiction of ASTM Committee D18 on Soil and Rock and is the direct responsibility of Subcommittee D18.07 on Identification and Classification of Soils.

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² Casagrande, A., "Classification and Identification of Soils," *Transactions, ASCE*, 1948, p. 901.

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

*A Summary of Changes section appears at the end of this standard



TABLE 1 Soil Classification Chart

Criteria for Assigning Group Symbols and Group Names Using Laboratory Tests ^A				Soil Classification	
				Group Symbol	Group Name ^B
COARSE-GRAINED SOILS	Gravels (More than 50 % of coarse fraction retained on No. 4 sieve)	Clean Gravels (Less than 5 % fines ^C)	Cu ≥ 4.0 and 1 ≤ Cc ≤ 3.0 ^D	GW	Well-graded gravel ^E
			Cu < 4.0 and/or [Cc < 1 or Cc > 3.0] ^D	GP	Poorly graded gravel ^E
		Gravels with Fines (More than 12 % fines ^C)	Fines classify as ML or MH	GM	Silty gravel ^{E,F,G}
			Fines classify as CL or CH	GC	Clayey gravel ^{E,F,G}
			Cu ≥ 6.0 and 1.0 ≤ Cc ≤ 3.0 ^D	SW	Well-graded sand ^I
			Cu < 6.0 and/or [Cc < 1.0 or Cc > 3.0] ^D	SP	Poorly graded sand ^I
FINE-GRAINED SOILS	Sands (50 % or more of coarse fraction passes No. 4 sieve)	Clean Sands (Less than 5 % fines ^H)	Fines classify as ML or MH	SM	Silty sand ^{F,G,I}
			Fines classify as CL or CH	SC	Clayey sand ^{F,G,I}
		Sands with Fines (More than 12 % fines ^H)	PI > 7 and plots on or above "A" line ^J	CL	Lean clay ^{K,L,M}
			PI < 4 or plots below "A" line ^J	ML	Silt ^{K,L,M}
			Liquid limit – oven dried Liquid limit – not dried < 0.75	OL	Organic clay ^{K,L,M,N} Organic silt ^{K,L,M,O}
			PI plots on or above "A" line	CH	Fat clay ^{K,L,M}
HIGHLY ORGANIC SOILS	Primarily organic matter, dark in color, and organic odor	Silt and Clays	PI plots below "A" line	MH	Elastic silt ^{K,L,M}
			Liquid limit – oven dried Liquid limit – not dried < 0.75	OH	Organic clay ^{K,L,M,P} Organic silt ^{K,L,M,Q}
		Silt and Clays	PI plots on or above "A" line	CH	Fat clay ^{K,L,M}
			PI plots below "A" line	MH	Elastic silt ^{K,L,M}
			Liquid limit – oven dried Liquid limit – not dried < 0.75	OH	Organic clay ^{K,L,M,P} Organic silt ^{K,L,M,Q}
			PI plots on or above "A" line	CH	Fat clay ^{K,L,M}

^ABased on the material passing the 3-in. (75-mm) sieve.

^BIf field sample contained cobbles or boulders, or both, add "with cobbles or boulders, or both" to group name.

^CGravels with 5 to 12 % fines require dual symbols:

GW-GM well-graded gravel with silt
GW-GC well-graded gravel with clay
GP-GM poorly graded gravel with silt
GP-GC poorly graded gravel with clay

$$^D Cu = D_{60} / D_{10} \quad Cc = \frac{(D_{30})^2}{D_{10} \times D_{60}}$$

^EIf soil contains ≥15 % sand, add "with sand" to group name.

^FIf fines classify as CL-ML, use dual symbol GC-GM, or SC-SM.

^GIf fines are organic, add "with organic fines" to group name.

^HSands with 5 to 12 % fines require dual symbols:

SW-SM well-graded sand with silt
SW-SC well-graded sand with clay
SP-SM poorly graded sand with silt
SP-SC poorly graded sand with clay

^IIf soil contains ≥15 % gravel, add "with gravel" to group name.

^JIf Atterberg limits plot in hatched area, soil is a CL-ML, silty clay.

^KIf soil contains 15 to <30 % plus No. 200, add "with sand" or "with gravel," whichever is predominant.

^LIf soil contains ≥30 % plus No. 200, predominantly sand, add "sandy" to group name.

^MIf soil contains ≥30 % plus No. 200, predominantly gravel, add "gravelly" to group name.

^NPI ≥ 4 and plots on or above "A" line.

^OPI < 4 or plots below "A" line.

^PPI plots on or above "A" line.

^QPI plots below "A" line.

C136 Test Method for Sieve Analysis of Fine and Coarse Aggregates

C702 Practice for Reducing Samples of Aggregate to Testing Size

D653 Terminology Relating to Soil, Rock, and Contained Fluids

D1140 Test Methods for Determining the Amount of Material Finer than 75-μm (No. 200) Sieve in Soils by Washing

D2216 Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass

D2488 Practice for Description and Identification of Soils (Visual-Manual Procedures)

D3740 Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

D4083 Practice for Description of Frozen Soils (Visual-Manual Procedure)

D4318 Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

D4427 Classification of Peat Samples by Laboratory Testing

D6913 Test Methods for Particle-Size Distribution (Grada- **tion) of Soils Using Sieve Analysis**

E11 Specification for Woven Wire Test Sieve Cloth and Test Sieves

3. Terminology

3.1 Definitions:

3.1.1 Except as listed below, all definitions are in accordance with Terminology **D653**.

NOTE 4—For particles retained on a 3-in. (75-mm) U.S. standard sieve, the following definitions are suggested:

Cobbles—particles of rock that will pass a 12-in. (300-mm) square opening and be retained on a 3-in. (75-mm) U.S. standard sieve, and

Boulders—particles of rock that will not pass a 12-in. (300-mm) square opening.

3.1.2 *clay*—soil passing a No. 200 (75- μ m) U.S. standard sieve that can be made to exhibit plasticity (putty-like properties) within a range of water contents and that exhibits considerable strength when air dry. For classification, a clay is a fine-grained soil, or the fine-grained portion of a soil, with a plasticity index equal to or greater than 4, and the plot of plasticity index versus liquid limit falls on or above the “A” line.

3.1.3 *gravel*—particles of rock that will pass a 3-in. (75-mm) sieve and be retained on a No. 4 (4.75-mm) U.S. standard sieve with the following subdivisions:

Coarse—passes 3-in. (75-mm) sieve and retained on $\frac{3}{4}$ -in. (19-mm) sieve, and

Fine—passes $\frac{3}{4}$ -in. (19-mm) sieve and retained on No. 4 (4.75-mm) sieve.

3.1.4 *organic clay*—a clay with sufficient organic content to influence the soil properties. For classification, an organic clay is a soil that would be classified as a clay except that its liquid limit value after oven drying is less than 75 % of its liquid limit value before oven drying.

3.1.5 *organic silt*—a silt with sufficient organic content to influence the soil properties. For classification, an organic silt is a soil that would be classified as a silt except that its liquid limit value after oven drying is less than 75 % of its liquid limit value before oven drying.

3.1.6 *peat*—a soil composed of vegetable tissue in various stages of decomposition usually with an organic odor, a dark-brown to black color, a spongy consistency, and a texture ranging from fibrous to amorphous.

3.1.7 *sand*—particles of rock that will pass a No. 4 (4.75-mm) sieve and be retained on a No. 200 (75- μ m) U.S. standard sieve with the following subdivisions:

Coarse—passes No. 4 (4.75-mm) sieve and retained on No. 10 (2.00-mm) sieve,

Medium—passes No. 10 (2.00-mm) sieve and retained on No. 40 (425- μ m) sieve, and

Fine—passes No. 40 (425- μ m) sieve and retained on No. 200 (75- μ m) sieve.

3.1.8 *silt*—soil passing a No. 200 (75- μ m) U.S. standard sieve that is nonplastic or very slightly plastic and that exhibits little or no strength when air dry. For classification, a silt is a fine-grained soil, or the fine-grained portion of a soil, with a plasticity index less than 4 or if the plot of plasticity index versus liquid limit falls below the “A” line.

3.2 Definitions of Terms Specific to This Standard:

3.2.1 *coefficient of curvature, C_c* —the ratio $(D_{30})^2/(D_{10} \times D_{60})$, where D_{60} , D_{30} , and D_{10} are the particle sizes corresponding to 60, 30, and 10 % finer on the cumulative particle-size distribution curve, respectively.

3.2.2 *coefficient of uniformity, C_u* —the ratio D_{60}/D_{10} , where D_{60} and D_{10} are the particle diameters corresponding to 60 and 10 % finer on the cumulative particle-size distribution curve, respectively.

4. Summary

4.1 As illustrated in **Table 1**, this classification system identifies three major soil divisions: coarse-grained soils, fine-grained soils, and highly organic soils. These three divisions are further subdivided into a total of 15 basic soil groups.

4.2 Based on the results of visual observations and prescribed laboratory tests, a soil is catalogued according to the basic soil groups, assigned a group symbol(s) and name, and thereby classified. The flow charts, **Figs. 1 and 2** for fine-grained soils, and **Fig. 3** for coarse-grained soils, can be used to assign the appropriate group symbol(s) and name.

5. Significance and Use

5.1 This standard classifies soils from any geographic location into categories representing the results of prescribed laboratory tests to determine the particle-size characteristics, the liquid limit, and the plasticity index.

5.2 The assigning of a group name and symbol(s) along with the descriptive information required in Practice **D2488** can be used to describe a soil to aid in the evaluation of its significant properties for engineering use.

5.3 The various groupings of this classification system have been devised to correlate in a general way with the engineering behavior of soils. This standard provides a useful first step in any field or laboratory investigation for geotechnical engineering purposes.

5.4 This standard may also be used as an aid in training personnel in the use of Practice **D2488**.

5.5 This standard may be used in combination with Practice **D4083** when working with frozen soils.

NOTE 5—Notwithstanding the statements on precision and bias contained in this standard: The precision of this test method is dependent on the competence of the personnel performing it and the suitability of the equipment and facilities used. Agencies that meet the criteria of Practice **D3740** are generally considered capable of competent and objective testing. Users of this test method are cautioned that compliance with Practice **D3740** does not in itself assure reliable testing. Reliable testing depends on several factors; Practice **D3740** provides a means for evaluating some of those factors.

6. Apparatus

6.1 A plasticity chart, similar to **Fig. 4**, and a cumulative particle-size distribution curve, similar to **Fig. 5**, are required.

NOTE 6—The “U” line shown on **Fig. 4** has been empirically determined to be the approximate “upper limit” for natural soils. It is a good check against erroneous data, and any test results that plot above or to the left of it should be verified.

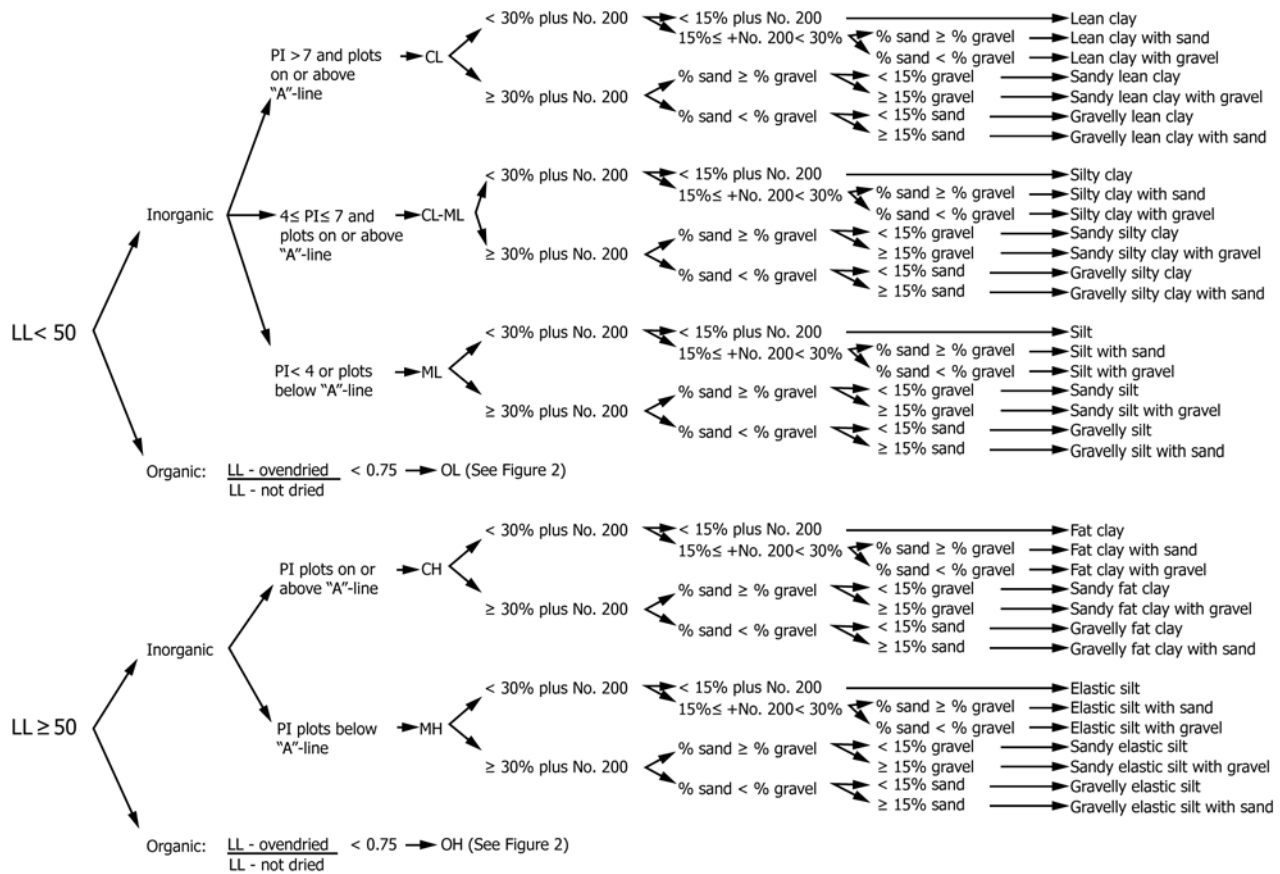


FIG. 1 Flow Chart for Classifying Fine-Grained Soil (50 % or More Passes No. 200 Sieve)

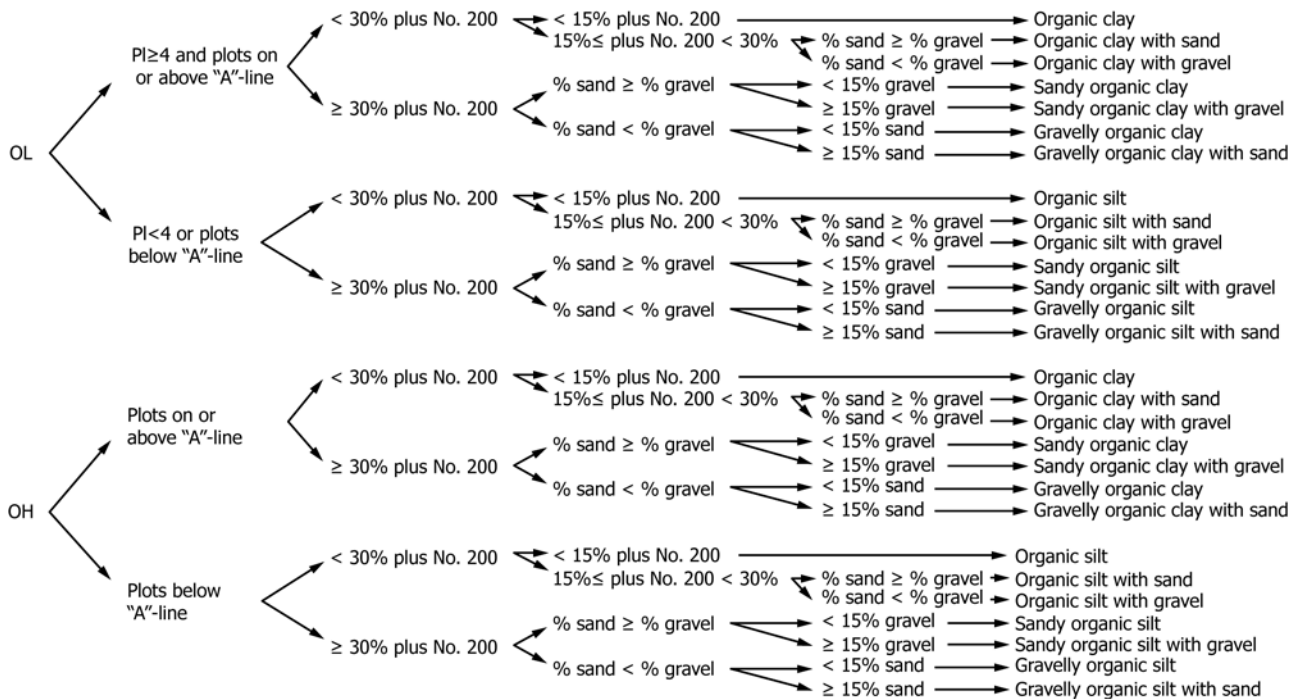


FIG. 2 Flow Chart for Classifying Organic Fine-Grained Soil (50 % or More Passes No. 200 Sieve)

7. Classification of Peat

7.1 A sample that is composed primarily of vegetative tissue in various stages of decomposition and has a fibrous to

amorphous texture, a dark-brown to black color, and an organic odor shall be classified as peat, PT, and not subjected to the classification procedures described hereafter.

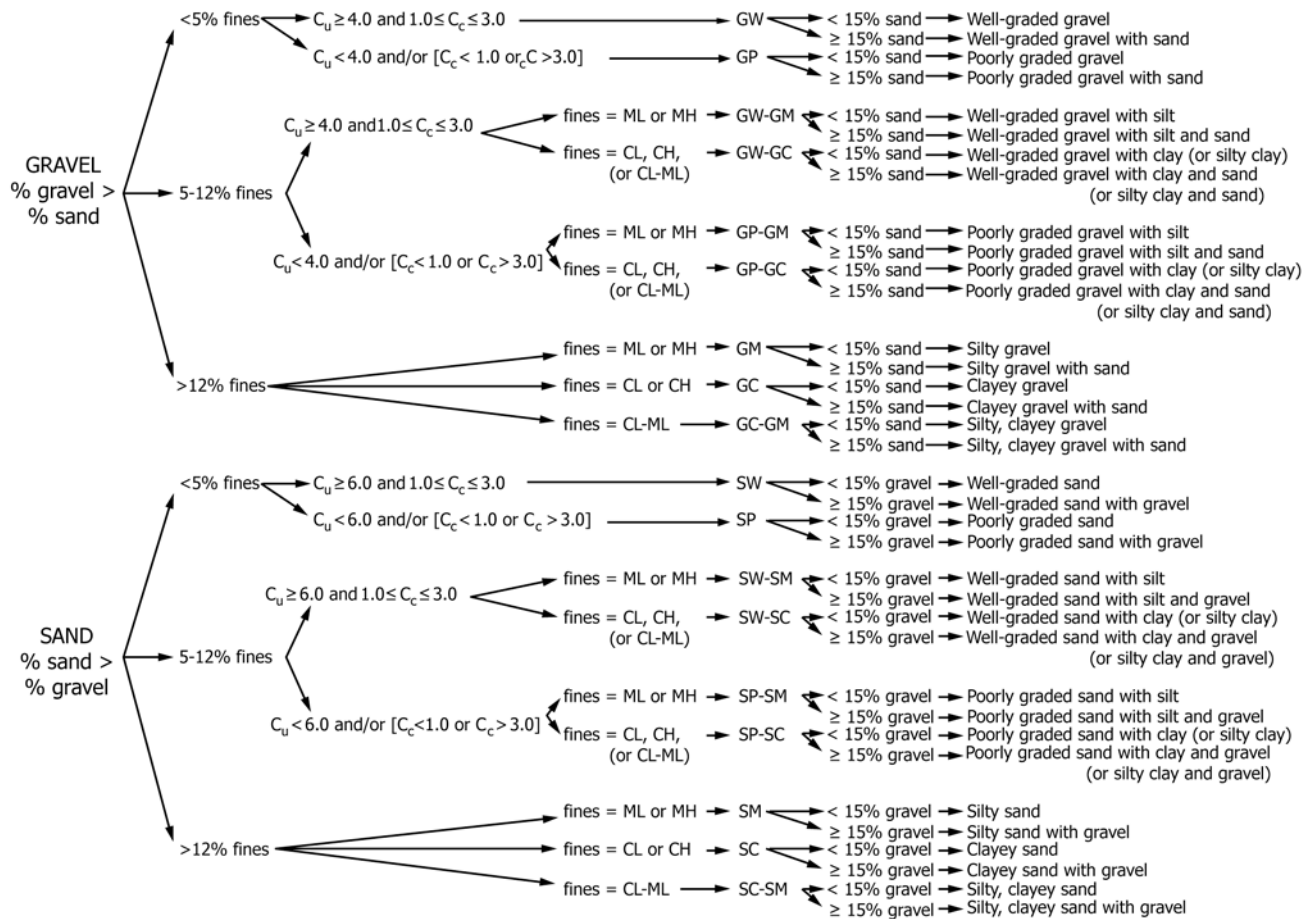


FIG. 3 Flow Chart for Classifying Coarse-Grained Soils (More Than 50 % Retained on No. 200 Sieve)

7.2 If desired, classification of type of peat can be performed in accordance with Classification D4427.

8. Preparation for Classification

8.1 Before a soil can be classified according to this standard, generally the particle-size distribution of the minus 3-in. (75-mm) material and the plasticity characteristics of the minus No. 40 (425- μ m) sieve material must be determined. See 8.4 for the specific required tests.

8.2 Although the test procedure used in determining the particle-size distribution or other considerations may require a hydrometer analysis of the material, a hydrometer analysis is not necessary for soil classification.

8.3 The percentage (by dry weight) of any plus 3-in. (75-mm) material must be determined and reported as auxiliary information. The maximum particle size shall be determined (measured or estimated) and reported as auxiliary information.

8.4 The tests results required for classification are as follows:

8.4.1 For soils estimated to contain less than 5 % fines, a plot of the cumulative particle-size distribution curve of the fraction coarser than the No. 200 (75- μ m) sieve is required. The cumulative particle-size distribution shall be performed in accordance with Test Method D6913, or with Test Method C136 after performing Test Method D1140 or Test Method

C117, if applicable. A semi-log plot of percent passing versus particle-size or sieve size/sieve number is plotted as shown in Fig. 5.

8.4.2 For soils estimated to contain 5 to 15 % fines, a cumulative particle-size distribution curve, as described in 8.4.1, is required, and the liquid limit and plasticity index in accordance with Test Method D4318 are required.

8.4.2.1 If sufficient material is not available to determine the liquid limit and plasticity index, the fines should be estimated to be either silty or clayey using the procedures described in Practice D2488 and so noted in the report.

8.4.3 For soils estimated to contain 15 % or more fines, a determination of the percent fines, percent sand, and percent gravel is required, and the liquid limit and plasticity index, as described in 8.4.2, are required. For soils estimated to contain 90 % fines or more, the percent fines, percent sand, and percent gravel may be estimated instead, using the procedures described in Practice D2488, and so noted in the report.

9. Preliminary Classification Procedure

9.1 Classify the soil as fine-grained if 50 % or more by dry mass of the test specimen passes the No. 200 (75- μ m) sieve and follow Section 10.

9.2 Classify the soil as coarse-grained if more than 50 % by dry mass of the test specimen is retained on the No. 200 (75- μ m) sieve and follow Section 11.

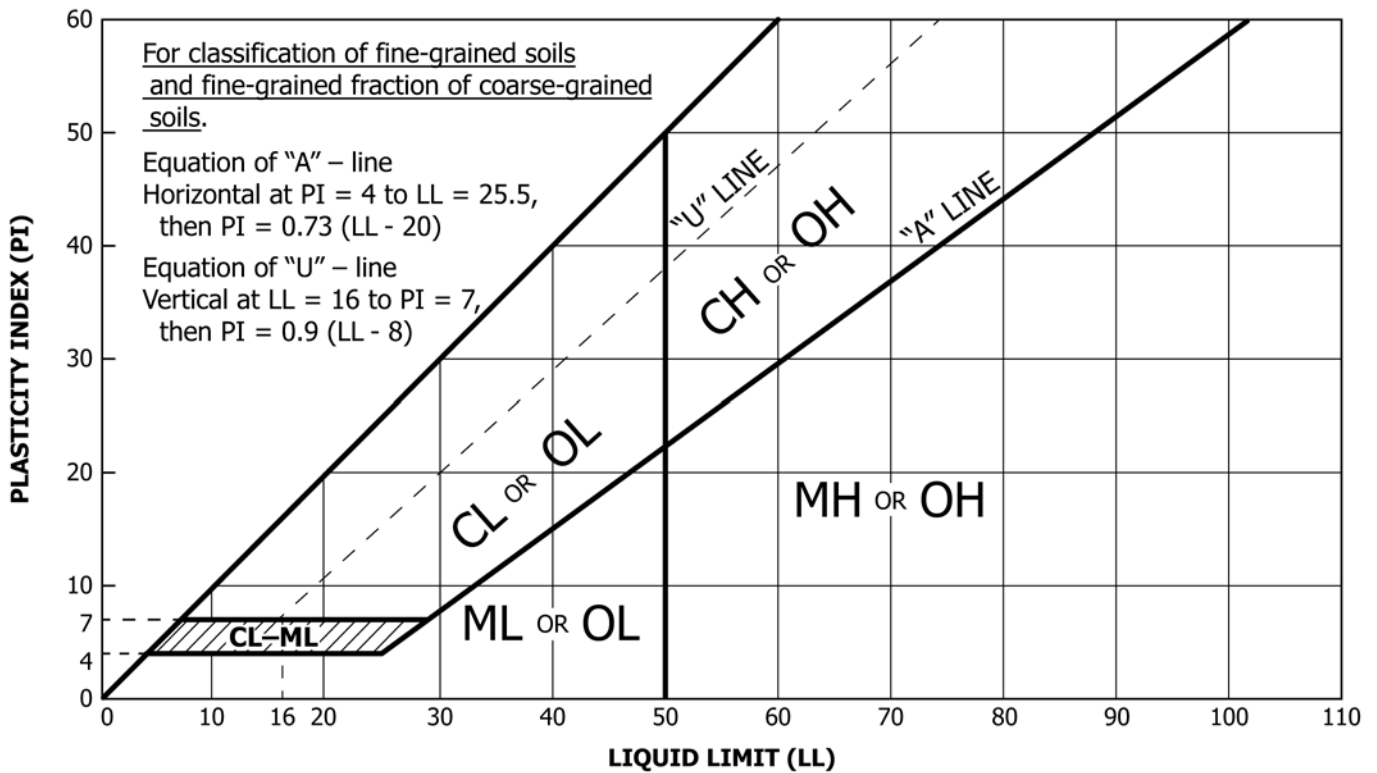
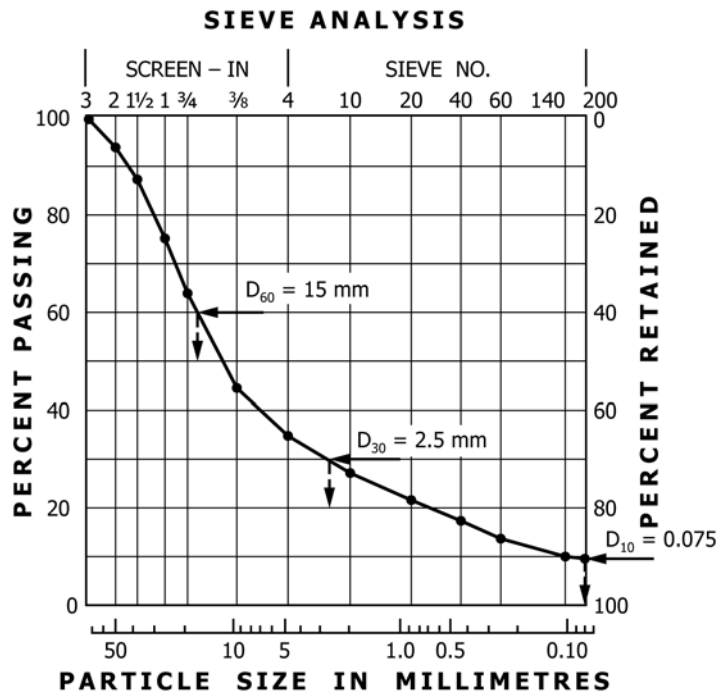


FIG. 4 Plasticity Chart



$$C_u = \frac{D_{60}}{D_{10}} = \frac{15}{0.075} = 200 \quad C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}} = \frac{(2.5)^2}{0.075 \times 15} = 5.6$$

FIG. 5 Cumulative Particle-Size Plot

10. Procedure for Classification of Fine-Grained Soils

(50 % or more by dry mass passing the No. 200 (75- μ m) sieve)

10.1 The soil is an inorganic clay if the position of the plasticity index versus liquid limit plot, Fig. 4, falls on or above the “A” line, the plasticity index is greater than 4, and the presence of organic matter does not influence the liquid limit as described in 10.3.2.

10.1.1 Classify the soil as a *lean clay*, CL, if the liquid limit is less than 50. See area identified as CL on Fig. 4.

10.1.2 Classify the soil as a *fat clay*, CH, if the liquid limit is 50 or greater. See area identified as CH on Fig. 4.

NOTE 7—In cases where the liquid limit exceeds 110 or the plasticity index exceeds 60, the plasticity chart may be expanded by maintaining the same scale on both axes and extending the “A” line at the indicated slope.

10.1.3 Classify the soil as a *silty clay*, CL-ML, if the position of the plasticity index versus liquid limit plot falls on or above the “A” line and the plasticity index is in the range of 4 to 7. See area identified as CL-ML on Fig. 4.

10.2 The soil is an inorganic silt if the position of the plasticity index versus liquid limit plot, Fig. 4, falls below the “A” line or the plasticity index is less than 4, and presence of organic matter does not influence the liquid limit as determined in 10.3.2.

NOTE 8—If performance of the Atterberg Limits test results in the soil being reported as nonplastic (D4318, 11.4), then assign the plasticity index a value of zero for classification purposes because the plotted result will fall below the “A” line.

10.2.1 Classify the soil as a *silt*, ML, if the liquid limit is less than 50. See area identified as ML on Fig. 4.

10.2.2 Classify the soil as an *elastic silt*, MH, if the liquid limit is 50 or greater. See area identified as MH on Fig. 4.

10.3 The soil is an organic silt or clay if organic matter is present in sufficient amounts to influence the liquid limit as determined in 10.3.2.

10.3.1 If the soil has a dark color and an organic odor when moist and warm, a second liquid limit test in accordance with D4318 shall be performed on a test specimen which has been oven dried at $110 \pm 5^\circ\text{C}$ to a constant mass.

10.3.2 The soil is an organic silt or organic clay if the liquid limit after oven drying is less than 75 % of the liquid limit of the original specimen determined before oven drying.

10.3.3 Classify the soil as an *organic silt* or *organic clay*, OL, if the liquid limit (not oven dried) is less than 50 %. Classify the soil as an *organic silt*, OL, if the plasticity index is less than 4, or the position of the plasticity index versus liquid limit plot falls below the “A” line. Classify the soil as an *organic clay*, OL, if the plasticity index is 4 or greater and the position of the plasticity index versus liquid limit plot falls on or above the “A” line. See area identified as OL (or CL-ML) on Fig. 4.

10.3.4 Classify the soil as an *organic clay* or *organic silt*, OH, if the liquid limit (not oven dried) is 50 or greater. Classify the soil as an *organic silt*, OH, if the position of the plasticity index versus liquid limit plot falls below the “A” line. Classify the soil as an *organic clay*, OH, if the position of the plasticity

index versus liquid-limit plot falls on or above the “A” line. See area identified as OH on Fig. 4.

10.4 If less than 30 % but 15 % or more of the test specimen is retained on the No. 200 (75- μ m) sieve, the words “with sand” or “with gravel” (whichever is predominant) shall be added to the group name. For example, lean clay with sand, CL; silt with gravel, ML. If the percent of sand is equal to the percent of gravel, use “with sand.”

10.5 If 30 % or more of the test specimen is retained on the No. 200 (75- μ m) sieve, the words “sandy” or “gravelly” shall be added to the group name. Add the word “sandy” if 30 % or more of the test specimen is retained on the No. 200 (75- μ m) sieve and the coarse-grained portion is predominantly sand. Add the word “gravelly” if 30 % or more of the test specimen is retained on the No. 200 (75- μ m) sieve and the coarse-grained portion is predominantly gravel. For example, sandy lean clay, CL; gravelly fat clay, CH; sandy silt, ML. If the percent of sand is equal to the percent of gravel, use “sandy.”

11. Procedure for Classification of Coarse-Grained Soils

(more than 50 % retained on the No. 200 (75- μ m) sieve)

11.1 Classify the soil as gravel if more than 50 % of the coarse fraction [plus No. 200 (75- μ m) sieve] is retained on the No. 4 (4.75-mm) sieve.

11.2 Classify the soil as sand if 50 % or more of the coarse fraction [plus No. 200 (75- μ m) sieve] passes the No. 4 (4.75-mm) sieve.

11.3 If 12 % or less of the test specimen passes the No. 200 (75- μ m) sieve, plot the cumulative particle-size distribution, Fig. 5, and compute the coefficient of uniformity, C_u , and coefficient of curvature, C_c , as given in Eqs 1 and 2.

$$C_u = D_{60}/D_{10} \quad (1)$$

$$C_c = (D_{30})^2/(D_{10} \times D_{60}) \quad (2)$$

where:

D_{10} , D_{30} , and D_{60} = the particle-size diameters corresponding to 10, 30, and 60 %, respectively, passing on the cumulative particle-size distribution curve, Fig. 5.

NOTE 9—It may be necessary to extrapolate the curve to obtain the D_{10} diameter.

11.3.1 If less than 5 % of the test specimen passes the No. 200 (75- μ m) sieve, classify the soil as a *well-graded gravel*, GW, or *well-graded sand*, SW, if C_u is greater than or equal to 4.0 for gravel or greater than 6.0 for sand, and C_c is at least 1.0 but not more than 3.0.

11.3.2 If less than 5 % of the test specimen passes the No. 200 (75- μ m) sieve, classify the soil as *poorly graded gravel*, GP, or *poorly graded sand*, SP, if either the C_u or the C_c criteria for well-graded soils are not satisfied.

11.4 If more than 12 % of the test specimen passes the No. 200 (75- μ m) sieve, the soil shall be considered a coarse-grained soil with fines. The fines are determined to be either clayey or silty based on the plasticity index versus liquid limit plot on Fig. 4. (See 8.4.2.1 if insufficient material available for testing).

11.4.1 Classify the soil as a *clayey gravel*, GC, or *clayey sand*, SC, if the fines are clayey, that is, the position of the plasticity index versus liquid limit plot, Fig. 4, falls on or above the “A” line and the plasticity index is greater than 7.

11.4.2 Classify the soil as a *silty gravel*, GM, or *silty sand*, SM, if the fines are silty, that is, the position of the plasticity index versus liquid limit plot, Fig. 4, falls below the “A” line or the plasticity index is less than 4.

11.4.3 If the fines plot as a silty clay, CL-ML, classify the soil as a *silty, clayey gravel*, GC-GM, if it is a gravel or a *silty, clayey sand*, SC-SM, if it is a sand.

11.5 If 5 to 12 % of the test specimen passes the No. 200 (75- μ m) sieve, give the soil a dual classification using two group symbols.

11.5.1 The first group symbol shall correspond to that for a gravel or sand having less than 5 % fines (GW, GP, SW, SP), and the second symbol shall correspond to a gravel or sand having more than 12 % fines (GC, GM, SC, SM).

11.5.2 The group name shall correspond to the first group symbol plus “with clay” or “with silt” to indicate the plasticity characteristics of the fines. For example, well-graded gravel with clay, GW-GC; poorly graded sand with silt, SP-SM (See 8.4.2.1 if sufficient material is not available for testing).

NOTE 10—If the fines plot as a *silty clay*, CL-ML, the second group symbol should be either GC or SC. For example, a poorly graded sand with 10 % fines, a liquid limit of 20, and a plasticity index of 6 would be classified as a poorly graded sand with silty clay, SP-SC.

11.6 If the specimen is predominantly sand or gravel but contains 15 % or more of the other coarse-grained constituent, the words “with gravel” or “with sand” shall be added to the group name. For example, poorly graded gravel with sand, clayey sand with gravel.

11.7 If the field sample contained any cobbles or boulders or both, the words “with cobbles,” or “with cobbles and boulders” shall be added to the group name. For example, silty gravel with cobbles, GM.

12. Report: Test Data Sheet(s)/Form(s)

12.1 A complete description shall include the group name, group symbol, and the results of the laboratory tests. The particle-size distribution shall be given in terms of percent of gravel, sand, and fines. The plot of the cumulative particle-size distribution curve shall be reported if used in classifying the soil. The results of the Atterberg limits tests shall be reported if used in classifying the soil. Report appropriate descriptive information according to the procedures in Practice D2488. A local or commercial name or geologic interpretation for the material may be added at the end of the descriptive information if identified as such.

12.2 The specimen preparation and test procedures used shall be referenced in the report.

NOTE 11—In some cases, because of a lack of space or a desire to convey more information with the group symbol alone, the optional abbreviated system presented in Appendix A3 may be more useful to indicate the soil classification symbol and name.

NOTE 12—Example: *Clayey Gravel with Sand and Cobbles* (GC)—46 % fine to coarse, hard, subrounded gravel; 30 % fine to coarse, hard, subrounded sand; 24 % clayey fines, LL = 38, PI = 19; weak reaction with HCl; original field sample had 4 % hard, subrounded cobbles; maximum dimension 150 mm.

In-Place Conditions—firm, homogeneous, dry, brown,

Geologic Interpretation—alluvial fan.

NOTE 13—Other examples of soil descriptions are given in Appendix X1.

13. Keywords

13.1 Atterberg limits; classification; clay; gradation; gravel; organic soils; particle-size distribution; sand; silt; soil classification; soil tests; USCS

APPENDIXES

(Nonmandatory Information)

X1. EXAMPLES OF DESCRIPTIONS USING SOIL CLASSIFICATION

X1.1 The following examples show how the information required in 12.1 can be reported. The appropriate descriptive information from Practice D2488 is included for illustrative purposes. The additional descriptive terms that would accompany the soil classification should be based on the intended use of the classification and the individual circumstances.

X1.1.1 *Well-Graded Gravel with Sand (GW)*—73 % fine to coarse, hard, subangular gravel; 23 % fine to coarse, hard, subangular sand; 4 % fines; C_c = 2.7, C_u = 12.4.

X1.1.2 *Silty Sand with Gravel (SM)*—61 % predominantly fine sand; 23 % silty fines, LL = 33, PI = 6; 16 % fine, hard, subrounded gravel; no reaction with HCl; (field sample smaller than recommended). In-Place Conditions—Firm, stratified and contains lenses of silt 1 to 2 in. thick, moist, brown to gray; in-place density = 106 lb/ft³ and in-place moisture = 9 %.

X1.1.3 *Organic Clay (OL)*—100 % fines, LL (not dried) = 32, LL (oven dried) = 21, PI (not dried) = 10; wet, dark brown, organic odor, weak reaction with HCl.

X1.1.4 *Silty Sand with Organic Fines (SM)*—74 % fine to coarse, hard, subangular reddish sand; 26 % organic and silty dark-brown fines, LL (not dried) = 37, LL (oven dried) = 26, PI (not dried) = 6, wet, weak reaction with HCl.

X1.1.5 *Poorly Graded Gravel with Silt, Sand, Cobbles and Boulders (GP-GM)*—78 % fine to coarse, hard, subrounded to subangular gravel; 16 % fine to coarse, hard, subrounded to subangular sand; 6 % silty (estimated) fines; moist, brown; no reaction with HCl; original field sample had 7 % hard, subrounded cobbles and 2 % hard, subrounded boulders with a maximum dimension of 18 in.

X2. USING SOIL CLASSIFICATION AS A DESCRIPTIVE SYSTEM FOR SHALE, CLAYSTONE, SHELLS, SLAG, CRUSHED ROCK, ETC.

X2.1 The group names and symbols used in this standard may be used as a descriptive system applied to materials that exist in situ as shale, claystone, sandstone, siltstone, mudstone, etc., but convert to soils after field or laboratory processing (crushing, slaking, etc.).

X2.2 Materials such as shells, crushed rock, slag, etc., should be identified as such. However, the procedures used in this standard for describing the particle size and plasticity characteristics may be used in the description of the material. If desired, a classification in accordance with this standard may be assigned to aid in describing the material.

X2.3 If a classification is used, the group symbol(s) and group names should be placed in quotation marks or noted with some type of distinguishing symbol. See examples.

X2.4 Examples of how soil classifications could be incorporated into a description system for materials that are not naturally occurring soils are as follows:

X2.4.1 *Shale Chunks*—Retrieved as 2- to 4-in. pieces of shale from power auger hole, dry, brown, no reaction with HCl.

After laboratory processing by slaking in water for 24 h, material classified as “Sandy Lean Clay (CL)”—61 % clayey fines, LL = 37, PI = 16; 33 % fine to medium sand; 6 % gravel-size pieces of shale.

X2.4.2 *Crushed Sandstone*—Product of commercial crushing operation; “Poorly Graded Sand with Silt (SP-SM)”—91 % fine to medium sand; 9 % silty (estimated) fines; dry, reddish-brown, strong reaction with HCl.

X2.4.3 *Broken Shells*—65 % gravel-size broken shells; 31 % sand and sand-size shell pieces; 4 % fines; Cc = 2.4, Cu = 1.9; would be classified as “Poorly Graded Gravel with Sand (GP)”.

X2.4.4 *Crushed Rock*—Processed gravel and cobbles from Pit No. 7; “Poorly Graded Gravel (GP)”—89 % fine, hard, angular gravel-size particles; 11 % coarse, hard, angular sand-size particles, dry, tan; no reaction with HCl; Cc = 2.4, Cu = 0.9.

X3. ABBREVIATED SOIL CLASSIFICATION SYMBOLS

X3.1 In some cases, because of lack of space, an abbreviated system may be useful to indicate the soil classification symbol and name. Examples of such cases would be graphical logs, databases, tables, etc.

X3.2 This abbreviated system is not a substitute for the full name and descriptive information but can be used in supplementary presentations when the complete description is referenced.

X3.3 The abbreviated system should consist of the soil classification symbol based on this standard with appropriate lower case letter prefixes and suffixes as:

Prefix	Suffix
s = sandy	s = with sand
g = gravelly	g = with gravel
	c = cobbles
	b = boulders

X3.4 The soil classification symbol is to be enclosed in parentheses. Some examples would be:

Group Symbol and Full Name	Abbreviated
CL, Sandy lean clay	s(CL)
SP-Sm, Poorly graded sand with silt and gravel	(SP-SM)g
GP, poorly graded gravel with sand, cobbles, and boulders	(GP)scb
ML, gravelly silt with sand and cobbles	g(ML)sc

SUMMARY OF CHANGES

Committee D18 has identified the location of selected changes to this practice since the last issue, D2487–11, that may impact the use of this practice. (December 15, 2017)

- (1) Deleted Appendix X3 (Wet Preparation Method) and Appendix X4 (Dry Preparation Method) since no tests are performed in this practice and the test method required thoroughly explain the wet and dry preparation method.
- (2) Changed instances of “weight” to “mass” throughout the standard.

- (3) Added Note 9 to provide guidance in instances where it is not possible to perform the plastic limit or the liquid limit test.
- (4) Removed the Precision and Bias section since it is not applicable to practices.



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